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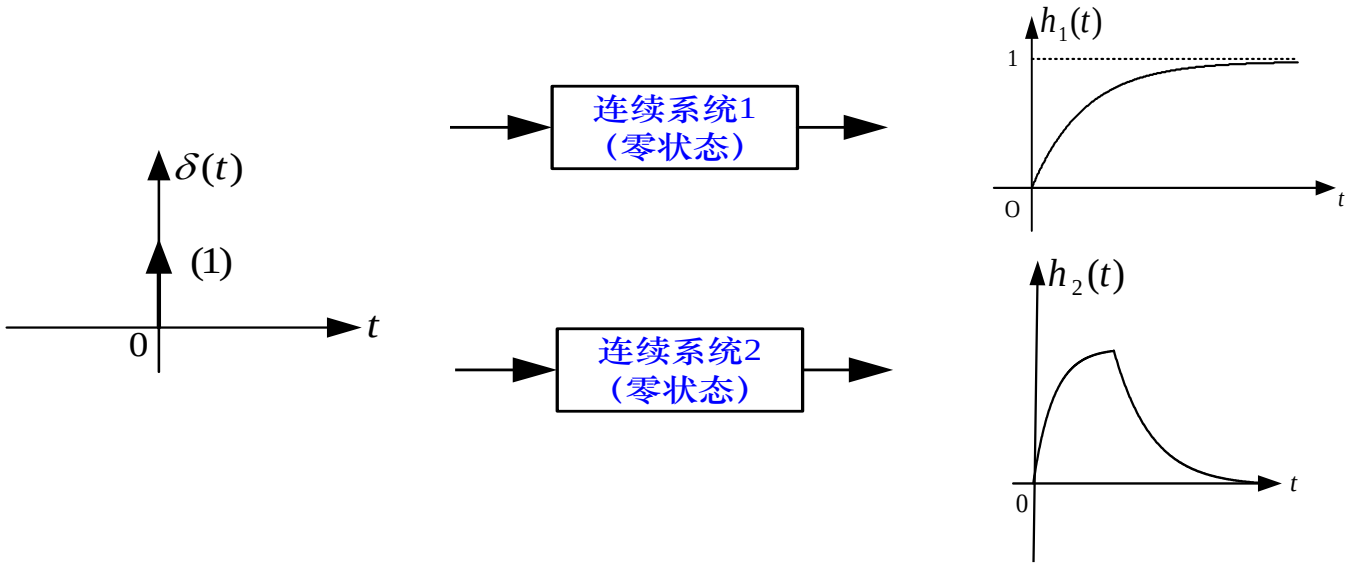
Impulse response represents system characteristics

- Impulse response of cascade system
- Impulse response of parallel system
- Impulse response and system causality
- Impulse response and system stability

Impulse response represents system characteristics



The impulse response $h(t)$ is also different for different systems

$h(t)$ is used to represent the time domain properties of continuous systems。

Impulse response represents system characteristics

Impulse response of several common systems

Distortion free transmission system: input: $d(t)$ $\Rightarrow$ $h(t) = K\delta(t - t_d)$

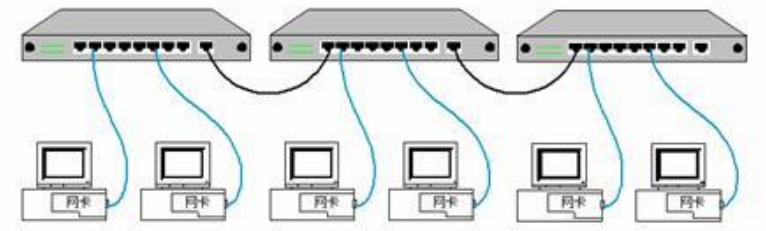
K is a normal number, t_d is the delay time after the input signal passes through the system.

Ideal integrator: input: $d(t)$ $\Rightarrow$ $h(t) = u(t)$

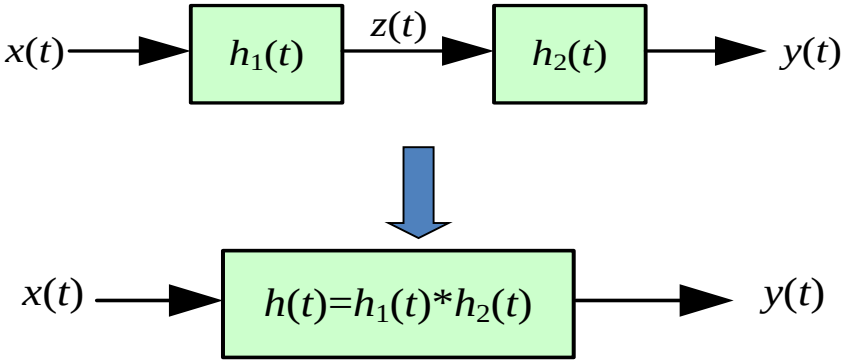
Ideal differentiator: input: $d(t)$ $\Rightarrow$ $h(t) = \delta'(t)$

1. Impulse response of cascade system

Cascade system



(Hub cascade)



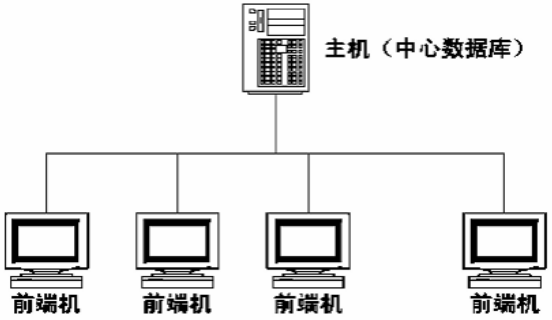
$$z(t) = x(t) * h_1(t)$$

$$y(t) = z(t) * h_2(t)$$

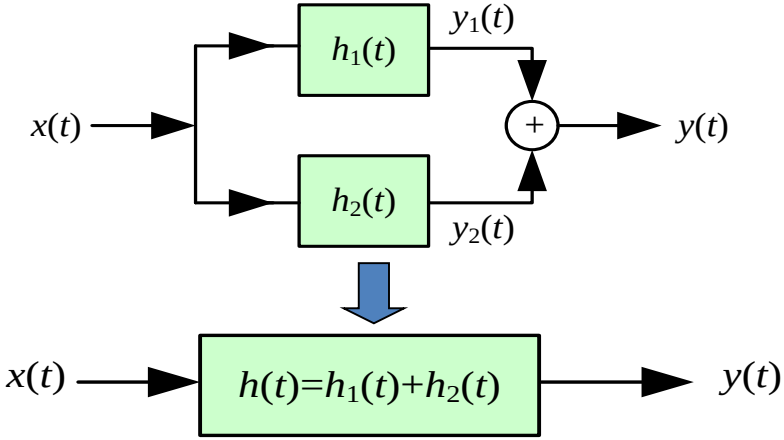
$$y(t) = x(t) * [\underbrace{h_1(t) * h_2(t)}_{h(t)}]$$

2. Impulse response of parallel system

System in parallel



(Computer parallel connection)



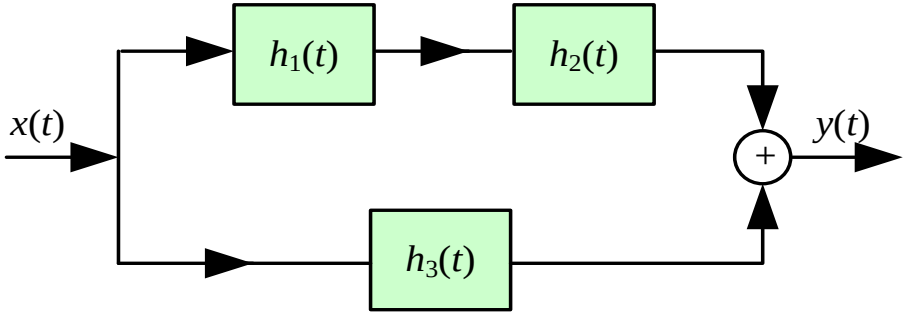
$$y_1(t) = x(t) * h_1(t)$$

$$y_2(t) = x(t) * h_2(t)$$

$$y(t) = x(t) * \underbrace{[h_1(t) + h_2(t)]}_{h(t)}$$

[Example] Figure out the impulse response $h(t)$ of the continuous system,

where $h_1(t) = e^{-3t}u(t)$, $h_2(t) = \delta(t-1)$, $h_3(t) = u(t)$



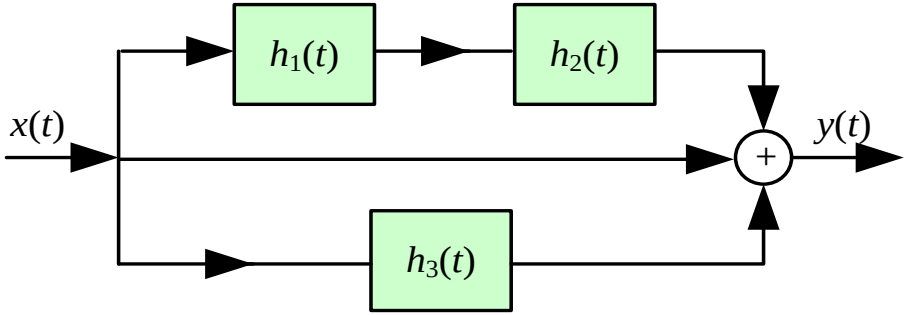
Solution: Sub-system $h_1(t)$ is cascaded with $h_2(t)$, and $h_3(t)$ branches are connected in parallel with $h_1(t)$ $h_2(t)$ branches

$$h(t) = h_1(t) * h_2(t) + h_3(t)$$

$$= \delta(t-1) * e^{-3t}u(t) + u(t)$$

$$= e^{-3(t-1)}u(t-1) + u(t)$$

[Example] Figure out the impulse response $h(t)$ of the continuous system, where $h_1(t) = e^{-3t}u(t)$, $h_2(t) = \delta(t-1)$, $h_3(t) = u(t)$。



Solution: $h(t) = h_1(t) * h_2(t) + h_3(t) + \delta(t)$

$$= \delta(t-1) * e^{-3t}u(t) + u(t) + \delta(t)$$

$$= e^{-3(t-1)}u(t-1) + u(t) + \delta(t)$$

3. Impulse response and system causality

Definition of causal system：

The output of the system at time t_0 is related only to the input signal at and before time t_0.

In other words, the output of the system is not ahead of the input.

For continuous LTI systems, it is a necessary and sufficient condition for causal systems

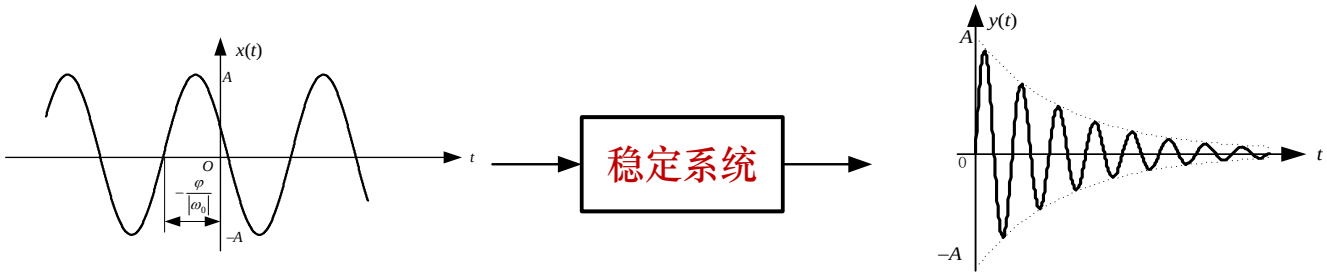
$$h(t) = 0, \quad t < 0$$

4. Impulse response and system stability

Stable system definition：

If the system has a bounded input it has a bounded output,

The system is called stable system.(BIBO stable)



For continuous LTI systems, it is a necessary and sufficient condition for stable system

$$\int_{-\infty}^{\infty} |h(\tau)| \mathrm{d}\tau = S < +\infty$$

[Example] The impulse response of a continuous LTI system is known to be $h(t) = e^{at}u(t)$, determine whether the system is causal and stable.

Solution: Because $h(t) = e^{at}u(t)$,

Meet $h(t) = 0, \quad t < 0$

Therefore, this system is a causal system.

Because $\int_{-\infty}^{\infty} |h(\tau)| d\tau = \int_{0}^{\infty} e^{a\tau} d\tau = \frac{1}{a} e^{a\tau}\Big|_{0}^{\infty}$

✓ When $a < 0$, $\int_{-\infty}^{\infty} |h(\tau)| d\tau = -\frac{1}{a}$ The system is stable.

✓ When $a \geq 0$, $\int_{-\infty}^{\infty} |h(\tau)| d\tau \to \infty$ The system is unstable.

Thanks

Some of the materials quoted in this course mainly talk about teachers' teaching accumulation for many years. They come from a variety of media and communication with colleagues, peers and friends. It is difficult to indicate the source one by one.